

Skill Studio

Where an agent's skills get measured, tried and trained

Internal sharing · September 2026

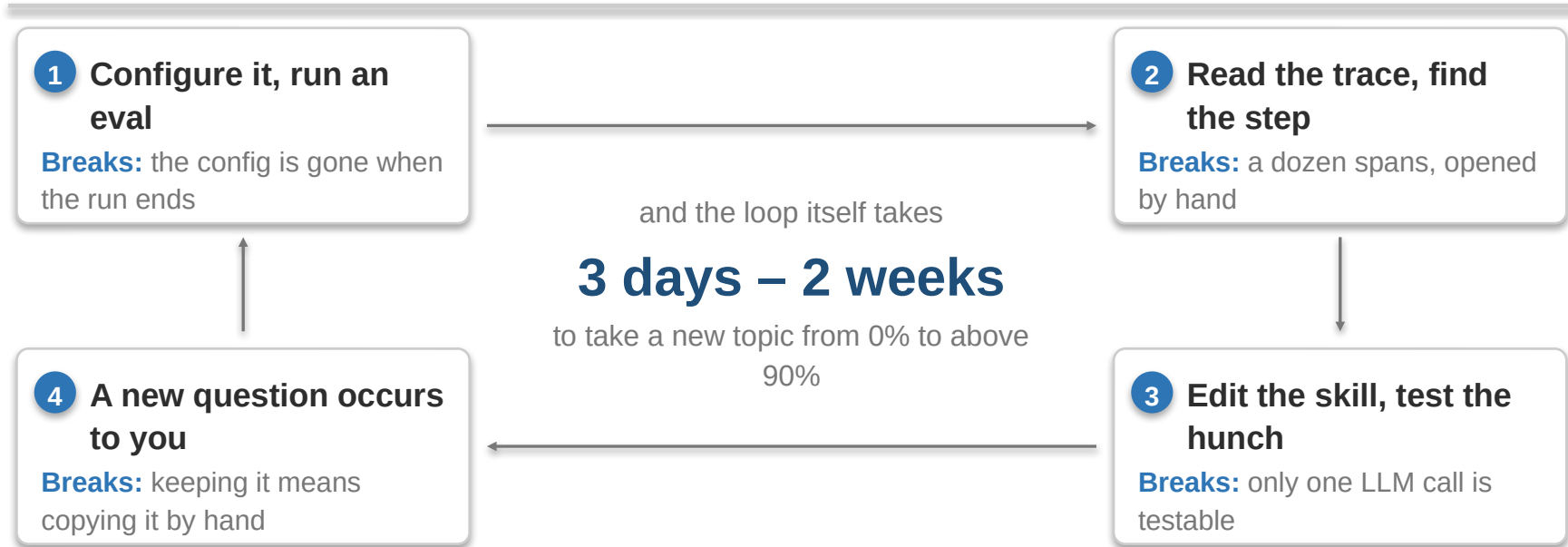
The studio makes them; the Skill Marketplace sells them

Today: the pain, the demo, and the algorithm behind Optimize

- | | | |
|----|--------------------------------------|-------|
| 01 | The pain | 8 min |
| | The loop, and where it breaks | |
| 02 | Demo | 5 min |
| | The real thing, in the browser | |
| 03 | The algorithm | 8 min |
| | A skill file as a trainable weight | |
| + | What's next | 2 min |
| | Cheaper optimization, shorter skills | |

Interrupt any time; the bigger questions keep until the end.

We run the same loop every day, and it breaks in four places



Langfuse is still the backbone our traces live in. What is missing is what it does not have and we cannot add to it: the error-localization engine, per-span information of our own, and skill optimization.

Skill Studio repairs all four breaks in one place

1 Config

Stored with every run, exactly as it ran

2 Traces

An LLM names the suspect span — a clue, not a verdict

3 Fixes

One question, an editable skill, one button

4 Questions

Drafted from the trace, then reviewed

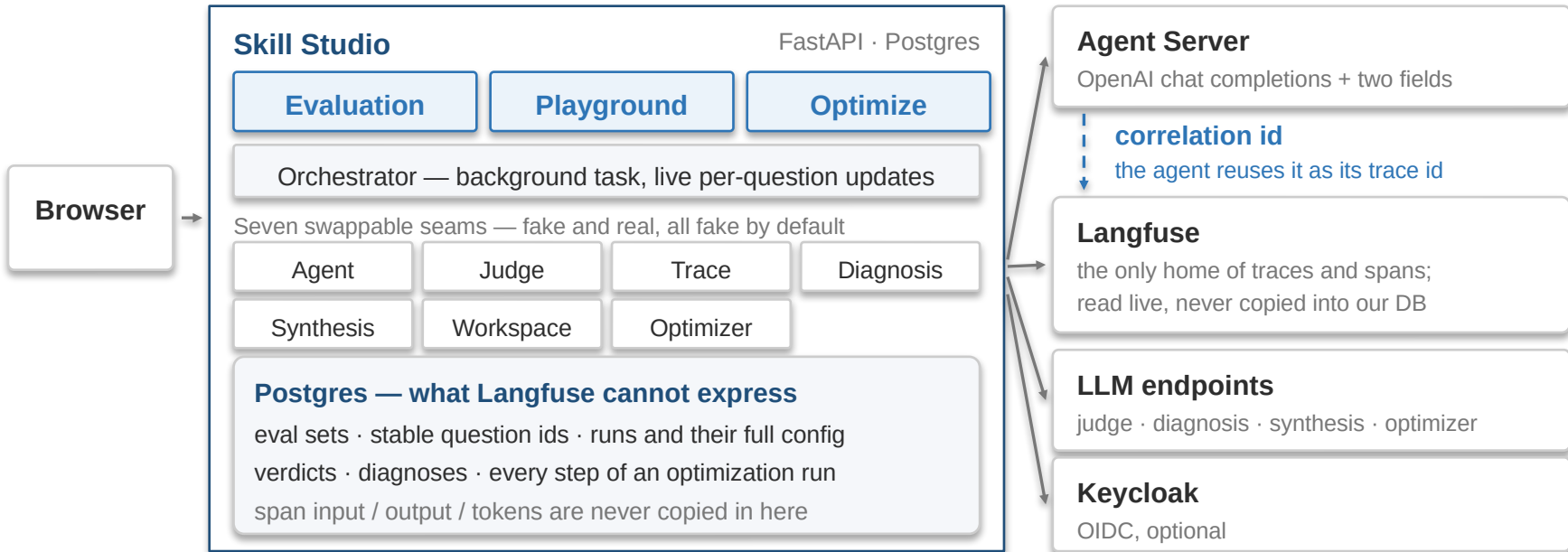
The loop itself takes 3 days to 2 weeks, so Optimize automates it

Train a skill like a model: epochs, steps, a learning rate, a validation gate.

- **isolated** trains the body; **routing** trains the description
- A chart per step, and a diff of what changed
- The output is a zip you install yourself

Runs cancel and resume; four early-stop conditions keep a bad night from burning to the end.

Langfuse owns traces; we own what Langfuse cannot express



All seven seams are fake by default: the whole system runs on nothing but Docker, and one real service can be brought up at a time.

Optimize turns weeks into one night, but not into zero people

BEFORE

3 days – 2 weeks

a person editing, re-running, reading results

NOW · isolated mode

overnight

press it before you leave, read it in the morning

- In use on several BOD and PPD topics
- The job moves to reviewing a skill that already passed validation
- Missing internal domain knowledge stalls accuracy near 80%

PART 2

Demo

- Run an eval — every question listed from the first second
- Open a failed one — the diagnosis jumps to the suspect span
- Carry it into the playground — edit the skill, ask again
- Shortlist it — draft the reasoning, make a new eval set
- Open an optimization run — the per-step chart and the diff

PART 3

SkillOpt

Treating a skill file as a trainable weight — the model is never touched

arXiv:2605.23904 (Microsoft) · our implementation is vendored from microsoft/SkillOpt, with our agent and our judge swapped in

Self-rewriting prompts overfit and forget, so SkillOpt treats the file as a weight

THE USUAL LOOP

- Overfits to whatever error just happened
- Bloats with one-off patches
- Quietly erases earlier lessons

SKILLOPT

- External, versioned, trainable state
- Bounded, controlled steps
- Kept only if it passes validation

That third failure is catastrophic forgetting — in plain text instead of in weights.

Every mechanism has a deep-learning counterpart

DEEP LEARNING

ITS TEXT-SPACE EQUIVALENT

Parameters W

The skill file `best_skill.md`

Gradient

Edit proposals from a minibatch

Learning rate

The edit budget, cosine $4 \rightarrow 2$

Validation checkpoint

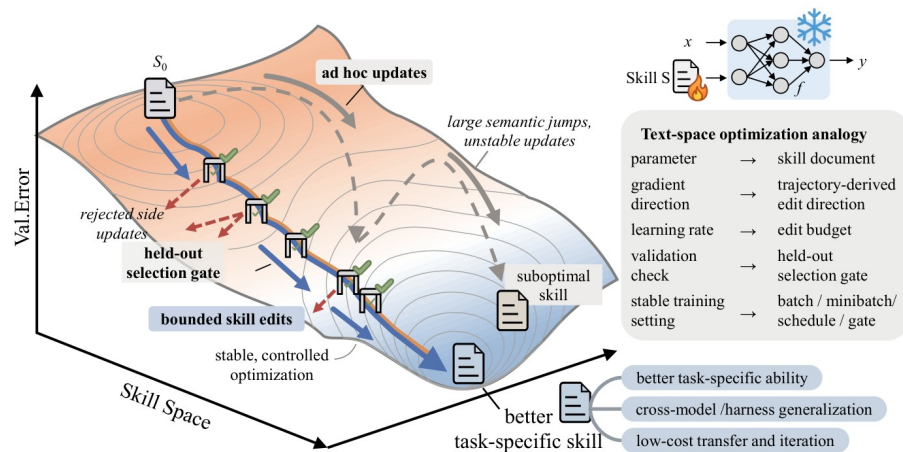
The blind `D_sel` gate, strictly greater

Momentum / EMA

The epoch-wise slow update

Each one exists to make the left-hand column work in text space.

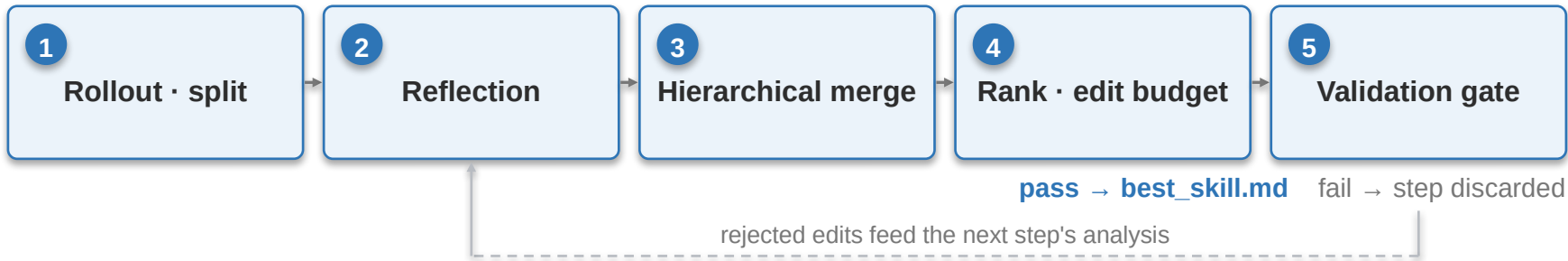
Separating the two models makes deployment cost zero



- **Target M:** frozen weights and prompt; only the skill file changes
- **Optimizer O:** offline only, reads trajectories, proposes edits
- When training ends, its API spend disappears

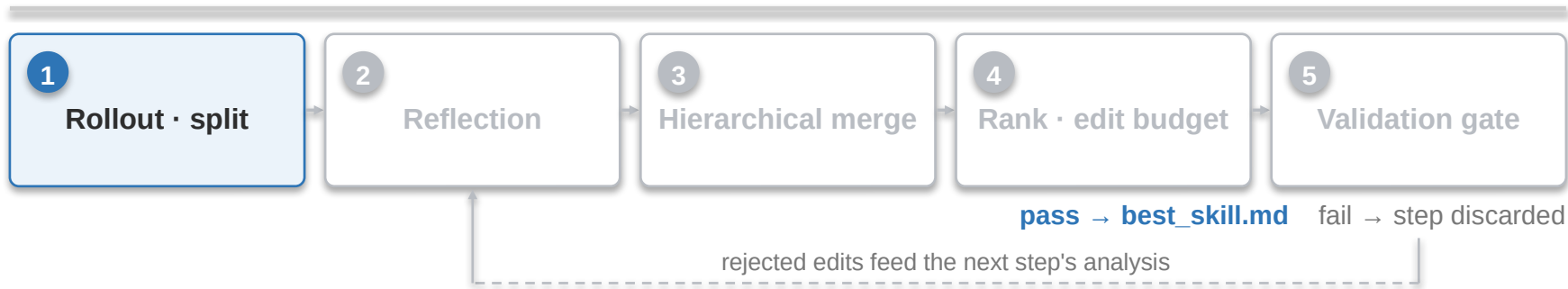
Figure 1. Source: SkillOpt, arXiv:2605.23904

One step is five stages, from evidence to gate



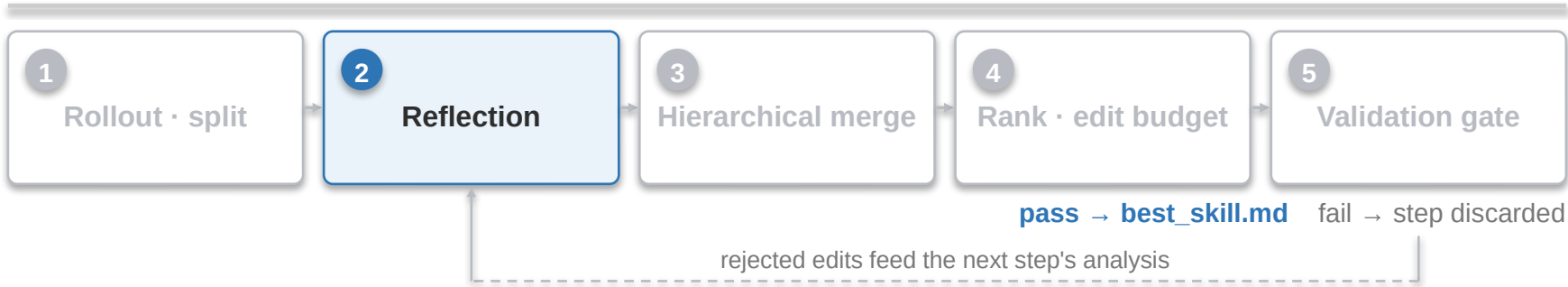
Gather the evidence, act, then earn it on an exam the optimizer has never seen.

A batch of 40 separates a systematic weakness from one-off noise



- React to one failure and you overfit to its noise
- 40 samples carry enough statistical weight
- Split by score: failures give fixes, successes give reinforcement

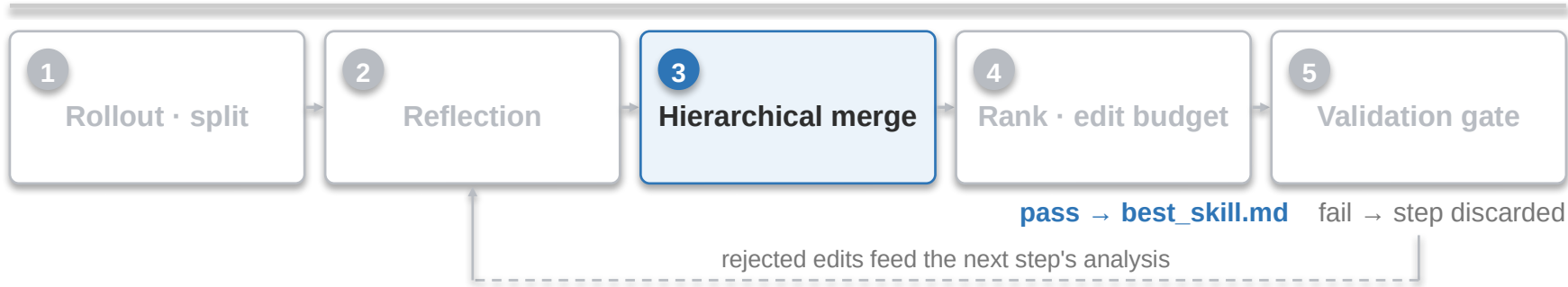
Minibatches of 8 find the failures that recur across tasks



- One trajectory overfits; a whole pool blows the context window
- Eight fits, and still forces comparison across tasks
- Each analyst returns atomic edits anchored to exact strings

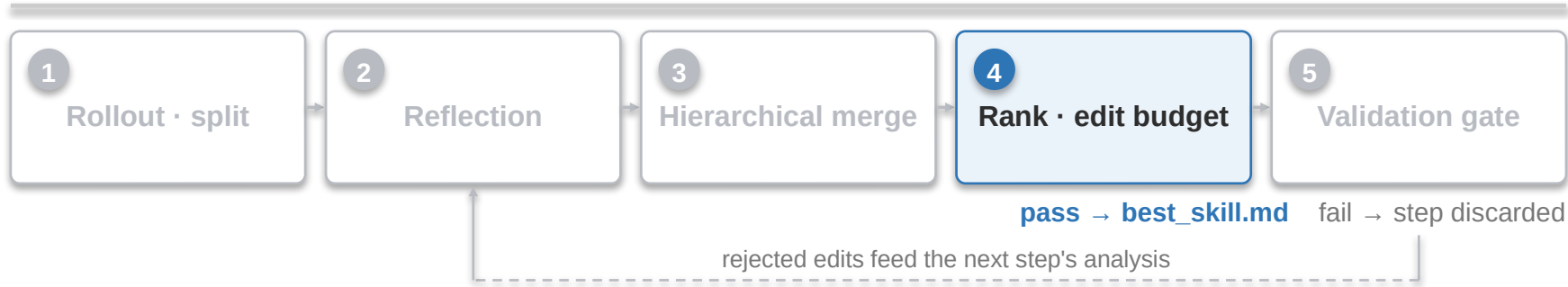
Up to 16 analyst calls run in parallel — Map, then Reduce.

Merging keeps the evidence, and fixing failures always wins



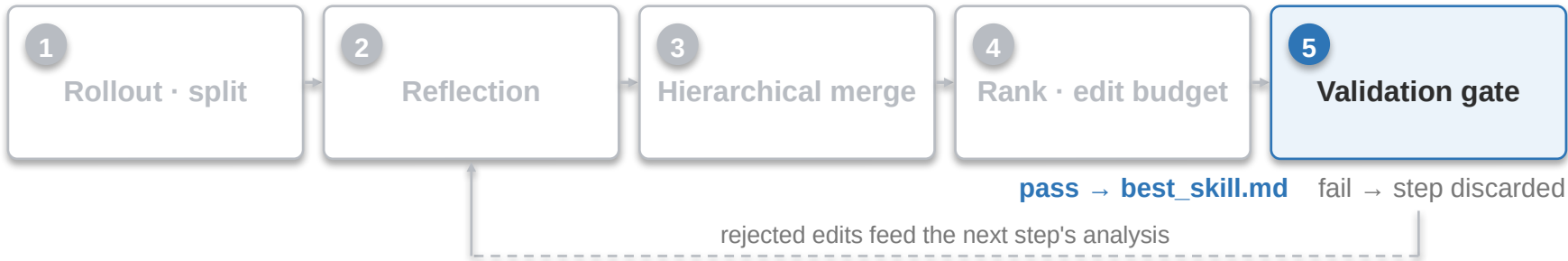
- Duplicates collapse to the most general wording, carrying support_count
- Conflicting edits are resolved here, not later

The edit budget is a learning rate, decaying 4 to 2



- Applying too many edits at once is too large a step
- Explore early, consolidate late
- Ranked by trajectories fixed, gap filled, generality, actionability

Only a strictly higher blind score is written to disk

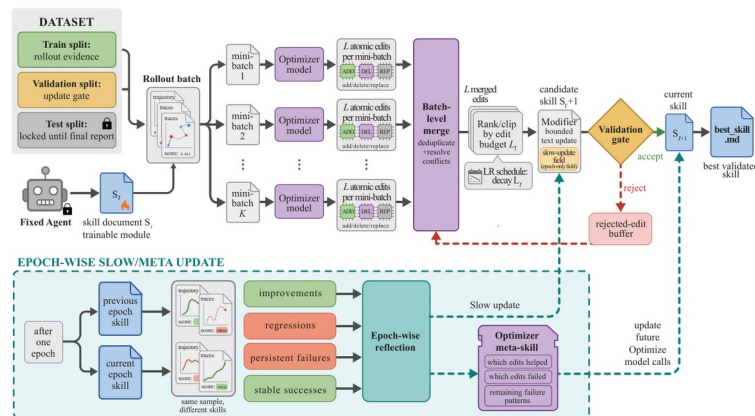


- The optimizer never sees D_{sel} at all
- A tie is discarded, not kept
- Rejected edits go into a buffer the next step reads

A hallucinated target string is skipped and logged, never fatal to the run.

An epoch-level loop catches what step-level edits erode

- Twenty tasks, old skill against new, one fixed exam
- Regressions expose edits that each passed their own gate
- The conclusion lands in a protected block — the momentum
- The meta skill teaches the optimizer, never the deployed file



What it learns reads like a senior engineer's review comment

SearchQA. “Infer the expected answer type from clue wording, then choose the shortest canonical entity supported by co-occurring distinctive evidence.”

SpreadsheetBench. “Inspect workbook structure and formulas, then write evaluated static values across the full requested target range instead of relying on Excel recalculation.”

OfficeQA. “Treat oracle parsed pages as primary evidence, lock table/date/unit context, and output exactly the requested rounded value without extra labels.”

DocVQA. “For tables, forms, charts, and legends, first bind the question to the exact visual row/header/field, then copy only the aligned answer span.”

LiveMathematicianBench. “In strongest-statement MCQs, rank choices by theorem strength and prefer a justified stronger-result option over true but weaker corollaries.”

ALFWorld. “Keep a horizon-aware visited/frontier ledger, diversify search after repeated same-type failures, and avoid revisiting the destination until holding the target.”

Our own developers write rules like these already — it takes them three days to two weeks.

Figure 4. Source: SkillOpt, arXiv:2605.23904

SkillOpt is best or tied-best in all 52 combinations

Model	Skill source	SourceQA	Spreadsheet	OfficeQA	DocVQA	LiveMath	ALFWorld
No harness / direct chat							
GPT-5.5	No skill	77.7	41.8	33.1	78.8	37.6	83.6
	Human skill	81.8 _{+1.1}	72.9 _{+30.1}	66.2 _{+33.8}	90.1 _{+11.3}	38.4 _{+8.9}	91.8 _{+3.2}
	LLM skill	80.9 _{+2.2}	43.2 _{+1.4}	51.7 _{+10.6}	80.6 _{+30.9}	40.0 _{+2.6}	93.3 _{+7.7}
	Trace2Skill	82.4 _{+1.7}	49.6 _{+1.8}	65.7 _{+23.6}	90.6 _{+11.8}	52.0 _{+14.4}	87.3 _{+3.7}
	TextGrad	81.4 _{+1.7}	41.1 _{+0.7}	42.0 _{+0.9}	87.2 _{+8.4}	49.2 _{+11.6}	82.8 _{+2.8}
	GEPA	81.6 _{+1.1}	72.6 _{+10.4}	63.9 _{+20.9}	89.1 _{+10.3}	43.2 _{+5.6}	85.8 _{+2.2}
	SkillOpt	87.3_{+1.6}	80.7_{+38.9}	72.1_{+10.0}	91.2_{+12.4}	60.9_{+20.3}	95.5_{+11.9}
GPT-5.4	No skill	76.9	41.8	30.0	77.6	36.5	75.4
	Human skill	80.1 _{+3.2}	59.3 _{+17.9}	59.3 _{+9.1}	88.5 _{+10.9}	41.6 _{+4.8}	74.6 _{+8.8}
	LLM skill	79.4 _{+2.5}	66.1 _{+4.7}	20.4 _{+26.6}	90.4 _{+12.8}	36.8 _{+6.0}	82.6 _{+4.2}
	Trace2Skill	81.2 _{+1.3}	53.2 _{+11.8}	51.2 _{+1.7}	89.3 _{+11.7}	40.0 _{+3.2}	79.1 _{+3.3}
	TextGrad	82.5 _{+1.6}	38.6 _{+2.8}	54.7 _{+4.7}	85.3 _{+7.7}	36.6 _{+0.2}	78.4 _{+3.0}
	GEPA	82.4 _{+1.5}	61.1 _{+18.7}	66.1 _{+10.4}	88.3 _{+10.7}	41.5 _{+4.4}	74.6 _{+4.8}
	SkillOpt	83.1_{+1.2}	62.5_{+11.1}	62.8_{+12.4}	91.2_{+13.6}	44.9_{+1.2}	91.0_{+13.6}
GPT-5.4-mini	No skill	75.9	36.1	22.1	71.4	14.7	73.1
	Human skill	77.2 _{+1.3}	42.9 _{+4.8}	43.3 _{+24.8}	85.0 _{+13.6}	28.8 _{+14.1}	36.7 _{+4.4}
	LLM skill	76.1 _{+1.2}	36.8 _{+0.7}	36.6 _{+13.5}	86.4 _{+15.0}	28.0 _{+13.3}	65.7 _{+7.4}
	Trace2Skill	78.6 _{+2.7}	40.7 _{+4.6}	20.9 _{+1.2}	85.5 _{+17.1}	32.8 _{+18.1}	52.8 _{+6.7}
	TextGrad	77.5 _{+1.6}	38.2 _{+2.3}	30.0 _{+1.9}	84.0 _{+12.8}	27.2 _{+12.5}	70.9 _{+1.2}
	GEPA	79.3 _{+3.5}	42.5 _{+6.4}	45.3 _{+12.2}	83.7 _{+12.3}	27.2 _{+12.5}	81.3 _{+3.2}
	SkillOpt	80.2_{+1.3}	47.5_{+11.4}	48.8_{+20.7}	90.8_{+10.3}	32.4_{+18.1}	85.8_{+12.7}
GPT-5.4-mono	No skill	55.8	23.5	16.3	30.8	23.2	34.3
	Human skill	66.1 _{+10.3}	41.8 _{+18.3}	46.5 _{+30.2}	73.5 _{+42.7}	16.8 _{+6.4}	29.9 _{+4.4}
	LLM skill	62.4 _{+6.6}	33.6 _{+10.1}	12.8 _{+3.2}	71.7 _{+10.9}	20.9 _{+3.2}	49.7 _{+11.4}
	Trace2Skill	69.9 _{+14.1}	35.4 _{+11.9}	16.3 _{+0.9}	72.3 _{+40.3}	25.6 _{+2.4}	58.2 _{+23.9}
	TextGrad	73.3 _{+17.6}	32.5 _{+8.0}	38.7 _{+22.4}	68.3 _{+17.5}	20.8 _{+2.4}	42.5 _{+4.2}
	GEPA	73.2 _{+17.4}	37.2 _{+13.7}	45.0 _{+18.7}	71.2 _{+10.4}	24.9 _{+1.6}	65.7 _{+14.4}
	SkillOpt	74.8_{+10.0}	42.5_{+19.0}	50.0_{+15.7}	80.2_{+10.4}	27.2_{+4.0}	69.4_{+15.3}
GPT-5.2	No skill	71.9	38.2	34.9	73.1	20.8	68.7
	Human skill	75.8 _{+1.9}	48.2 _{+10.0}	55.2 _{+10.3}	89.0 _{+15.9}	23.2 _{+2.4}	96.7 _{+10.0}
	LLM skill	73.7 _{+1.8}	38.6 _{+0.4}	14.0 _{+0.9}	87.7 _{+14.6}	25.9 _{+4.8}	68.7 _{+6.0}
	Trace2Skill	79.2 _{+1.9}	43.9 _{+5.7}	43.0 _{+8.7}	87.7 _{+14.6}	28.8 _{+6.0}	70.9 _{+1.2}
	TextGrad	82.6 _{+10.7}	32.1 _{+4.1}	51.1 _{+10.2}	87.2 _{+14.3}	22.4 _{+1.6}	70.1 _{+1.4}
	GEPA	82.7 _{+10.6}	34.8 _{+10.1}	53.4 _{+14.3}	89.0 _{+15.9}	25.9 _{+4.8}	82.8 _{+11.1}
	SkillOpt	83.1_{+11.2}	37.1_{+18.0}	36.4_{+11.5}	89.6_{+16.5}	36.9_{+15.3}	85.1_{+16.8}
Qwen3.5-4B	No skill	68.1	9.3	14.5	86.9	22.4	30.6
	Human skill	66.3 _{+1.8}	18.4 _{+7.1}	22.2 _{+4.2}	87.8 _{+10.9}	18.4 _{+4.8}	28.4 _{+2.2}
	LLM skill	65.0 _{+3.1}	10.4 _{+1.3}	20.9 _{+6.1}	88.0 _{+11.9}	28.5 _{+4.8}	55.2 _{+20.0}
	Trace2Skill	68.5 _{+0.4}	12.3 _{+0.0}	16.5 _{+1.8}	88.0 _{+1.3}	27.2 _{+4.8}	64.9 _{+14.3}
	TextGrad	60.7 _{+1.4}	13.9 _{+1.6}	20.9 _{+6.1}	85.6 _{+1.8}	10.6 _{+1.1}	55.7 _{+10.1}
	GEPA	65.6 _{+0.5}	16.9 _{+1.6}	22.7 _{+4.2}	85.1 _{+1.8}	28.5 _{+4.8}	60.4 _{+20.8}
	SkillOpt	71.2_{+1.1}	23.9_{+14.6}	28.7_{+1.2}	90.8_{+1.3}	32.8_{+20.6}	81.9_{+10.7}
Qwen3.5B-32B-A3B	No skill	72.7	38.2	45.9	87.6	31.2	50.7
	Human skill	74.1 _{+1.4}	44.3 _{+6.1}	41.9 _{+4.0}	88.4 _{+0.8}	29.6 _{+1.6}	44.8 _{+14.8}
	LLM skill	72.6 _{+1.1}	42.9 _{+1.7}	36.3 _{+1.6}	88.4 _{+0.8}	24.9 _{+4.4}	60.4 _{+1.7}
	Trace2Skill	75.4 _{+2.7}	33.2 _{+3.8}	32.0 _{+13.9}	90.4 _{+1.8}	29.6 _{+1.6}	70.9 _{+1.2}
	TextGrad	76.3 _{+3.7}	22.9 _{+15.3}	33.7 _{+13.2}	84.5 _{+3.3}	7.2 _{+24.0}	67.9 _{+4.2}
	GEPA	75.8 _{+1.1}	45.4 _{+7.2}	43.6 _{+3.2}	89.0 _{+0.4}	31.2 _{+1.6}	73.2 _{+1.2}
	SkillOpt	80.3_{+1.6}	47.5_{+8.3}	47.1_{+1.2}	91.4_{+1.8}	41.8_{+10.4}	82.1_{+22.4}
Codex harness							
GPT-5.5	No skill	81.8	27.5	38.3	87.2	35.2	—
	Human skill	85.1 _{+2.3}	90.7 _{+23.2}	40.0 _{+1.7}	88.8 _{+1.6}	48.8 _{+13.6}	—
	LLM skill	83.4 _{+1.6}	25.0 _{+2.8}	34.1 _{+3.3}	89.8 _{+1.6}	45.0 _{+10.4}	—
	EvoSkill	61.4 _{+20.4}	67.5 _{+40.0}	42.4 _{+1.1}	89.3 _{+1.1}	63.2 _{+28.0}	—
	SkillOpt	87.3_{+1.5}	85.9_{+17.5}	51.1_{+12.8}	92.2_{+1.0}	78.4_{+13.2}	—
Claude Code harness							
GPT-5.5	No skill	81.9	22.1	57.6	86.6	40.8	—
	Human skill	83.7 _{+1.8}	37.1 _{+15.0}	66.3 _{+3.7}	88.0 _{+1.4}	44.0 _{+3.2}	—
	LLM skill	82.4 _{+1.3}	27.1 _{+13.0}	56.4 _{+1.2}	89.6 _{+1.0}	44.9 _{+4.0}	—
	EvoSkill	84.0 _{+1.1}	73.0 _{+32.9}	70.3 _{+12.7}	87.2 _{+0.6}	52.0 _{+11.2}	—
	SkillOpt	85.9_{+1.0}	86.4_{+18.3}	71.4_{+13.0}	90.1_{+1.5}	56.5_{+15.7}	—

- +23.5 direct chat, +24.8 Codex, +19.1 Claude Code
- One to four accepted edits per benchmark
- Ahead of TextGrad, GEPA and EvoSkill

Table 1. Source: SkillOpt, arXiv:2605.23904

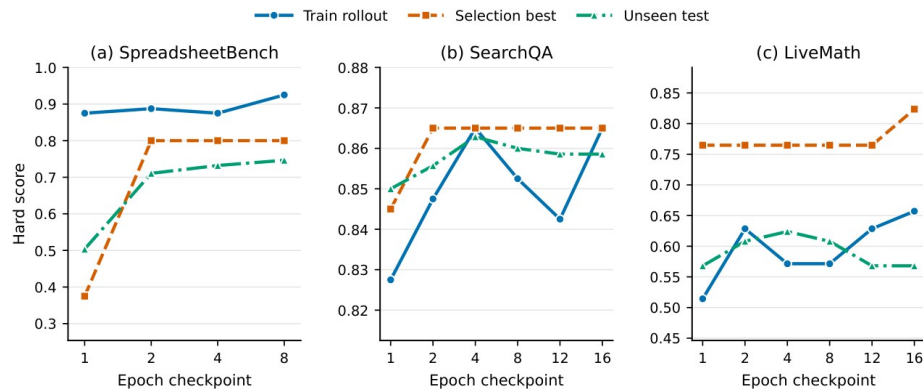
A trained skill transfers, and beats training in the target harness

(a) Cross-model transfer					
Source model	Target model	Benchmark	Baseline	Direct	Transferred
GPT-5.4	GPT-5.4	SpreadsheetBench	41.4	62.5	—
	GPT-5.4-mini		36.1	47.5	45.5 ^{+9.4}
	GPT-5.4-nano		23.5	42.5	26.5 ^{+3.0}
GPT-5.4	GPT-5.4	LiveMath	36.8	44.0	—
	GPT-5.4-mini		14.7	32.8	19.2 ^{+4.5}
	GPT-5.4-nano		23.2	27.2	28.8 ^{+5.6}
(b) Cross-harness transfer					
Source harness	Target harness	Benchmark	Baseline	Direct	Transferred
Codex	Claude Code	LiveMath	40.8	56.5	42.4 ^{+1.6}
Claude Code	Codex		35.2	78.4	48.0 ^{+12.8}
Codex	Claude Code	SpreadsheetBench	22.1	80.4	81.8 ^{+59.7}
Claude Code	Codex		27.5	85.0	71.1 ^{+43.6}
(c) Cross-benchmark transfer					
Source benchmark	Target benchmark	Model	Baseline	Direct	Transferred
OlympiadBench	Omni-MATH	GPT-5.4	56.6	—	60.3 ^{+3.7}
		GPT-5.4-mini	34.8	—	36.6 ^{+1.8}
		GPT-5.4-nano	38.8	—	40.1 ^{+1.3}

- Codex → Claude Code:
22.1 → 81.8, above the
80.4 trained there
- No transferred cell fell
below the target's own no-
skill baseline

Table 4. Source: SkillOpt, arXiv:2605.23904

The gate selects for generalization, not for overfit



- The validation peak lines up with the test peak
- Demote the optimizer to the target model and 56–74% of the gain survives

Figure 3. Source: SkillOpt, arXiv:2605.23904

Routing needed a different algorithm, and the oscillation proved it

isolated

a skill's body — thousands of words

routing

a skill's description — one line of
YAML

- Edits are mutually exclusive; merge picks without the evidence
- Failure and success are one boundary; split, they oscillate
- Routing decides before the agent acts

Two directions could make this cheaper and shorter

Training cost today: 0.6–1.1M tokens per point on short tasks, 38–46M on long-context ones.

CHEAPER

SkillOpt-Lite [arXiv:2607.03451](#)

- A minimal pipeline; +8.8 and +25.4 on LiveMath

NPO [arXiv:2608.27266](#)

- Matches GEPA with fewer rollouts

SHORTER

SkillZip [arXiv:2608.11079](#)

- Evaluation-free compression of a skill that grew by appending

Adjacent work, shaping what we feed it: Skill Self-Play [arXiv:2607.22529](#) · SkillWiki [arXiv:2606.16523](#)

CONCLUSIONS

- 1 Every break in the loop is worth repairing
- 2 A skill can be trained like a weight, with no model change
- 3 The optimizer does not replace people; people run the last mile

SEED=1 ./scripts/dev.sh brings the whole thing up — all seven seams are fake by default

Design record: [backend/docs/spec.md](#) · Routing: [backend/docs/routing-optimization.md](#)

References

SkillOpt. arXiv:2605.23904, Microsoft. — Figures 1–4 and Tables 1, 3, 4 reproduced in this deck.

microsoft/SkillOpt — the implementation vendored into Optimize.

SkillOpt-Lite: Better and Faster Agent Self-evolution via One Line of Vibe. arXiv:2607.03451.

Naive Prompt Optimization: Rethinking the Need for Complex Prompt Search. arXiv:2608.27266.

Skill Self-Play: Pushing the Frontier of LLM Capability with Co-Evolving Skills. arXiv:2607.22529.

SkillWiki: A Living Knowledge Infrastructure for Agent Skills. arXiv:2606.16523.

SkillZip: Evaluation-Free Skill Compression for Self-Evolving Agents by Discovering Reusable Structure. arXiv:2608.11079.

Langfuse — langfuse.com. Trace and span storage; read through its public API.

Internal: backend/docs/spec.md · routing-optimization.md · agent-server-api.md

Your agent server implements two endpoints

```
POST {chat endpoint}      an ordinary OpenAI chat completions call
  "skill_studio": {
    "trace_data": { "trace_id": "..." }  ← use this as the Langfuse trace id
    (required)
    "timeout_s": 120                      ← this call's budget
    "skills": { "billing/SKILL.md": "..." } ← whole file set, this call only
  }
GET  {skills endpoint}    → {"skills": {path: text}, "version": "..."}  (optional)
```

- Without the correlation id the platform cannot find the trace it just caused
- skills replaces the directory rather than patching it, so {} and absent differ

Every mechanism pays for itself

Component	Setting	SearchQA	SpreadsheetBench	LiveMath
Learning-rate form	lr=4 (default)	87.1	77.5	61.3
	dynamic lr	85.8	71.8	54.0
	without lr	84.6	75.7	57.3
Rejected buffer	with rejected buffer	87.1	77.5	61.3
	without rejected buffer	85.5	72.9	58.9
Slow/meta update	meta skill and slow update	87.1	77.5	61.3
	without meta skill	85.1	75.7	58.1
	without meta skill and slow update	86.3	55.0	59.7

- Each row removes one component and re-measures
- The learning-rate form, the rejected buffer and the epoch-wise update all earn their place

Table 3. Source: SkillOpt, arXiv:2605.23904

What we deliberately did not build

- Skills are never written back — the output is a zip
- Answer hard-coding has three defences
- Slow update and meta skill ship disabled
- Diagnosis accuracy itself is still unmeasured
- A tool or base-model error cannot be fixed by a skill